

GL Towers Dashboard

Functional Logic Specification

Version 1.0

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1. Purpose

This document is the authoritative functional logic specification for the GL Towers Dashboard Version 1.0. It describes the complete operational behavior of every dashboard component: sensor processing, health evaluation, alert generation, recommendation logic, state management, and user interaction handling.

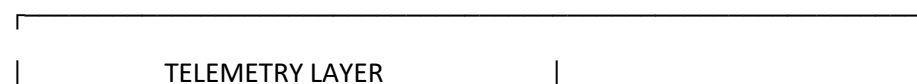
The primary audience is the embedded firmware team responsible for recreating the dashboard behavior inside the embedded touchscreen display. No web source code is required in addition to this document.

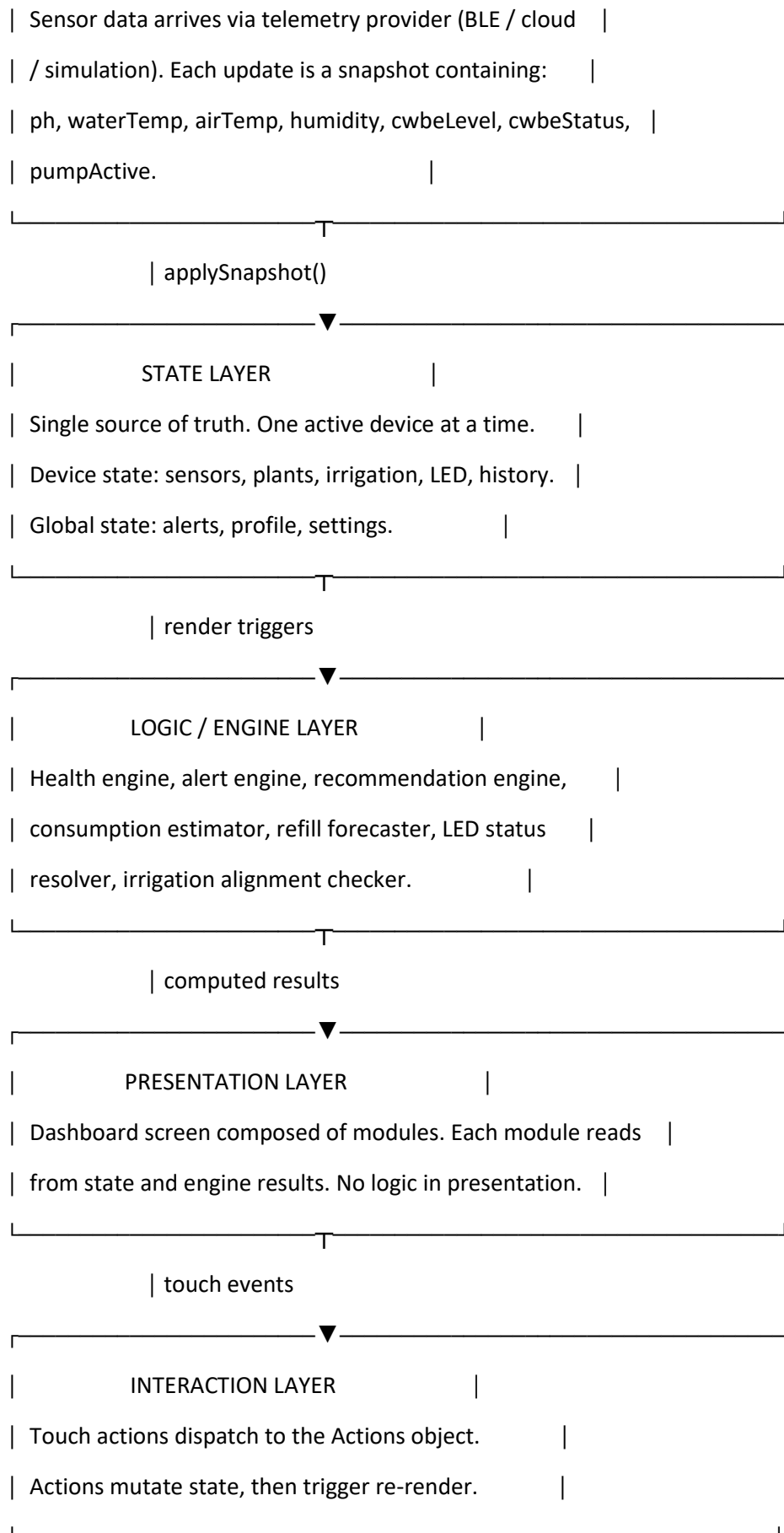
2. System Architecture Overview

2.1 Component Topology

The GL Towers Dashboard is organized around five conceptual layers:

...





...

2.2 Device Data Model

Each managed tower (device) maintains the following persistent state:

Field	Type	Description
-----	-----	-----
`id`	string	Unique device identifier (generated at creation)
`hardwareId`	string	Hardware MAC / BLE device ID / cloud token (null until paired)
`name`	string	Human-readable tower name
`room`	string	Optional room label
`tankLiters`	float	Virtual reservoir fill level in liters
`irrigationSaved`	boolean	True when irrigation has been configured and saved
`sensors`	object	Current telemetry snapshot (transient, not persisted)
`savedPlants`	object	Confirmed crop configuration
`savedLed`	object	Confirmed LED schedule
`savedIrrigation`	object	Confirmed irrigation program
`telemetryHistory`	object	Rolling sample array (up to 480 samples)
`metadata`	object	Device ID, firmware version, connection status

2.3 Default Device State

When a device is created with no prior configuration, the following defaults apply:

Parameter	Default Value
-----	-----
pH	6.2
Water Temperature	22.0 °C
Air Temperature	23.5 °C
Humidity	70 %

CWBE Level	75 %	
CWBE Status	ok	
Pump	inactive	
LED preset	Standard (06:00–22:00, 80%)	
Substrate	Hydroponic	
Growth stage	Vegetative	
Irrigation ON	1 min	
Irrigation PAUSE	90 min	
Tank fill	6.0 L (nominal capacity)	

3. Dashboard Philosophy

3.1 Data-Driven Display

Every value displayed on the dashboard is derived directly from the sensor telemetry snapshot and the user-configured parameters. No values are estimated or interpolated for display purposes unless explicitly stated in this document. The display always reflects the most recent valid snapshot.

3.2 Single Active Device

The dashboard displays data for one active device at a time. Multi-device selection is available at the navigation level, but all modules show only the active device.

3.3 Degraded Mode Operation

When telemetry data is unavailable or stale, the dashboard does not display incorrect data. Instead, modules enter defined degraded states and display explicit indicators rather than potentially incorrect values.

3.4 Rule-Based Intelligence

All health evaluations, alerts, and recommendations are produced by a deterministic rule engine. No machine learning or probabilistic inference is used in Dashboard V1. The firmware can reproduce all outputs exactly by implementing the same rules with the same threshold values.

3.5 Draft vs. Saved State

For LED and irrigation configuration, the dashboard maintains a two-layer state: a draft (being edited) and a saved (confirmed) configuration. All operational logic (health engine, alert engine, refill forecast) operates exclusively on saved configuration. Draft changes have no effect on health evaluations until explicitly saved by the operator.

4. Dashboard Refresh Cycle

4.1 Telemetry Tick

The dashboard refresh is event-driven from the telemetry provider. In the current implementation, the telemetry provider fires a new sensor snapshot at the following configurable intervals:

Speed Setting	Tick Interval
----- -----	
Slow	30,000 ms
Normal	15,000 ms
Fast	5,000 ms

Default speed: **Normal (15,000 ms)**.

4.2 Actions on Each Telemetry Tick

When a new telemetry snapshot is received, the following sequence executes:

1. Merge snapshot into active device sensors: `{ ph, waterTemp, airTemp, humidity, waterLevel, pumpActive, lastUpdate: now }`
2. Update device connection metadata (compute `connectionStatus` from timestamp age).
3. Record a history sample if minimum sample interval (10,000 ms) has elapsed since the previous sample.
4. Re-render the Home screen if it is currently active.
5. Update the Agronomist telemetry strip (health summary band at top of agronomist panel).
6. Run the notification generation engine (throttled to a minimum interval of 8,000 ms between runs).
7. Persist device state to storage every 4 ticks (~60 s at normal speed).

4.3 History Recording

The telemetry history is a rolling circular buffer with the following parameters:

Parameter	Value
-----	-----
Maximum samples	480
Minimum sample interval	10,000 ms
Maximum stored coverage	~2 hours at 15 s ticks

Each history sample contains: `{ t, ph, waterTemp, airTemp, humidity, waterLevel, tankLiters, stabilityScore }`.

Old samples are removed (FIFO) when the buffer reaches 480 entries. The history is persisted to storage every 4 recorded samples.

4.4 Screen Refresh on Navigation

Navigation actions trigger a full screen re-render. Sensor values displayed are always the last received telemetry snapshot at the moment of render.

5. Sensor Inputs

5.1 pH

Attribute	Value	
-----	-----	
Sensor type	pH probe in nutrient solution	
Unit	pH units (dimensionless, range 0–14)	
Telemetry field	`sensors.ph`	
Refresh	Every telemetry tick (see Section 4.1)	
Display precision	2 decimal places (e.g., 6.20)	

Validation:

Any numeric value in the range 0.0–14.0 is accepted. Values outside this range indicate a sensor fault and must be treated as missing data.

Missing data behavior:

If ``sensors.lastUpdate` == null`, the sensor tile displays "–" and the health engine does not evaluate pH rules.

Display behavior:

The pH value is displayed in the Sensor Grid on the Home screen. The tile background top-bar is colored green (brand color). The value is shown in large numeric type with no unit suffix.

Insight mapping (getSensorInsight):

pH Value Range	Status Code	Display Text	
-----	-----	-----	
< 5.5	warn	Critical low	
5.5 – < 5.8	low	Slightly low	
5.8 – 6.5	ok	Stable	

> 6.5 – ≤ 7.2	low	Slightly high	
> 7.2 – ≤ 7.5	warn	High	
> 7.5	warn	Critical high	

5.2 Water Temperature

Attribute	Value	
-----	-----	
Sensor type	Temperature probe in nutrient solution	
Unit	Degrees Celsius (°C)	
Telemetry field	`sensors.waterTemp`	
Refresh	Every telemetry tick	
Display precision	1 decimal place (e.g., 21.5°C)	

Validation:

Any numeric value. Physical plausibility range: 0–50 °C. Values outside plausible range indicate sensor fault.

Missing data behavior:

If `sensors.lastUpdate === null`, displays "-" and health engine rules for water temperature do not fire.

Display behavior:

Displayed in Sensor Grid. Tile background top-bar is blue. Unit suffix "°" is shown.

Insight mapping (getSensorInsight):

Water Temp Range	Status Code	Display Text	
-----	-----	-----	
< 16 °C	warn	Too cold	

16 – < 18 °C	low	Slightly cool	
18 – < 23 °C	ok	Optimal	
23 – < 25 °C	low	Slightly warm	
25 – < 27 °C	low	Warm	
≥ 27 °C	warn	Too warm — low O ₂ risk	

5.3 Air Temperature

Attribute	Value	
-----	-----	
Sensor type	Temperature sensor inside tower enclosure	
Unit	Degrees Celsius (°C)	
Telemetry field	`sensors.airTemp`	
Refresh	Every telemetry tick	
Display precision	1 decimal place	

Validation:

Physical plausibility range: 0–50 °C.

Missing data behavior:

If `sensors.lastUpdate` === null, displays "—" and health engine rules do not fire.

Display behavior:

Displayed in Sensor Grid. Tile background top-bar is amber/orange. Unit suffix "°".

Insight mapping (getSensorInsight):

Air Temp Range	Status Code	Display Text	
-----	-----	-----	

< 12 °C	warn	Critical cold	
12 – < 16 °C	low	Slightly cool	
16 – < 24 °C	ok	Good for growth	
24 – < 26 °C	low	Slightly warm	
26 – < 28 °C	low	Warm	
≥ 28 °C	warn	Heat stress risk	

5.4 Relative Humidity

Attribute	Value	
-----	-----	
Sensor type	Humidity sensor inside tower enclosure	
Unit	Percent relative humidity (% RH)	
Telemetry field	`sensors.humidity`	
Refresh	Every telemetry tick	
Display precision	Integer (e.g., 90%)	

****Validation:****

Valid range: 0–100 %. Values outside this range indicate sensor fault.

****Missing data behavior:****

If `sensors.lastUpdate === null`, displays "-" and health engine rules do not fire.

****Display behavior:****

Displayed in Sensor Grid. Tile background top-bar is purple. Unit suffix "%".

****Stage-dependent insight mapping:****

Humidity targets are growth-stage-dependent. The active stage is taken from `savedIrrigation.stage`.

Stage-specific target ranges:

Growth Stage	RH Target Min	RH Target Max
Germination	80 %	96 %
Seedling	70 %	96 %
Vegetative	60 %	96 %
Mature	50 %	96 %

****Insight mapping with active stage (getSensorInsight):****

Condition	Status	Display Text
RH < stage.min - 10	warn	Very low internal RH
stage.min - 10 ≤ RH < stage.min	low	RH below stability target
stage.min ≤ RH ≤ stage.max	ok	Stable
stage.max < RH ≤ 99	low	High internal RH
RH > 99	warn	Near-saturated internal RH

****Insight mapping without stage (fallback):****

Condition	Status	Display Text
RH < 65	warn	Very low internal RH
65 ≤ RH < 80	low	RH below stability target
80 ≤ RH ≤ 99	ok	Stable
RH > 99	warn	Near-saturated internal RH

6. Health Status Engine

The Health Status Engine is the primary classification system that assigns a single operational status to the monitored tower based on all available sensor data.

6.1 Status Levels

The engine produces one of five possible states:

Status	Meaning
-----	-----
Excellent	All monitored parameters within their optimal ranges. No active rules.
Good	System operating within acceptable bounds. Only informational (info) rules active.
Attention	One or more parameters outside preferred ranges. Warning-level rules active.
Critical	One or more parameters require immediate operator action. Critical-level rules active.
No Data	No telemetry snapshot has been received yet (<code>`lastUpdate === null`</code>). No rules evaluated.

6.2 Rule Evaluation

The engine maintains a library of agronomist rules (see Section 10 for the full rule library). On each telemetry update:

1. Each rule's ``condition`` function is evaluated against the current device state.
2. If a condition guard ``dev.sensors.lastUpdate === null`` is true, the rule returns false and does not fire.
3. All rules whose conditions evaluate to true are collected as ``firedRules``.
4. ``firedRules`` are sorted by ascending ``priority`` (lowest number = highest priority evaluated first).

6.3 Health Score Computation

The health score is computed from the set of fired rules:

****Step 1 — Penalty deduction:****

Start with a base score of ****100****.

For each fired rule, subtract the penalty associated with its severity:

Severity	Penalty Deducted
-----	-----
critical	- 40
warning	- 15
info	- 5

Score is clamped to a minimum of ****0****.

****Step 2 — Status from score:****

Score Range	Status from Score
-----	-----
≥ 85	excellent
65 – 84	good
40 – 64	attention
0 – 39	critical

****Step 3 — Status from rules:****

Independently of the score:

Fired Rule Severities Present	Status from Rules
-----	-----
Any critical rule fired	critical
Any warning rule fired	attention

Any info rule fired	good	
No rules fired	excellent	

****Step 4 — Final status (worst-case merge):****

The final status is the worse of `statusFromScore` and `statusFromRules`, where severity rank is: excellent (0) < good (1) < attention (2) < critical (3).

...

```
final_status = max_rank(statusFromScore, statusFromRules)
```

...

6.4 No-Data State

When `sensors.lastUpdate === null`, the health engine returns no status. The dashboard displays a "No Data" indicator rather than any health classification.

6.5 State Transitions

Previous State	New Condition	New State
-----	-----	-----
Any	lastUpdate = null	No Data
No Data	First snapshot received	Evaluated
Excellent	Any info rule fires	Good
Excellent/Good	Any warning rule fires	Attention
Any	Any critical rule fires	Critical
Critical	All critical rules clear	Attention or lower
Attention	All warning rules clear	Good or Excellent
Good	All info rules clear	Excellent

7. Health Score

7.1 Purpose

The health score is a single integer value in the range 0–100 that summarizes the operational status of the tower in a format suitable for display as a gauge or numeric indicator. It is derived from the same rule evaluation as the Health Status Engine.

7.2 Computation

The health score is computed as described in Section 6.3, Steps 1–2.

Starting value: ****100****

Deductions applied per fired rule (cumulative).

Floor: ****0**** (cannot go below zero).

7.3 Score-to-Status Mapping

Score Range	Status Label	Dashboard Color
-----	-----	-----
85 – 100	Excellent	Green
65 – 84	Good	Green (lighter)
40 – 64	Attention	Amber
0 – 39	Critical	Red

7.4 Display Behavior

The health score is displayed as a numeric value (0–100) accompanied by the status label. The display color follows the color column in the table above.

A progress bar or arc gauge may optionally visualize the score. The fill level corresponds directly to the score value (0 = empty, 100 = full).

7.5 Limitations

- The health score reflects the current snapshot only. A score of 100 with no history does not imply long-term stability.
- Multiple simultaneous issues stack penalties. A tower with two warning-level issues scores lower than a tower with one warning-level issue.
- The score does not incorporate trend information (rising or falling sensor values) in V1. Trend information is used only in the Agronomist panel text.

7.6 Future Evolution

The health score architecture is designed to accommodate additional penalty sources in future versions, including trend penalties, plant-profile deviation scoring, and CWBE efficiency penalties. No structural changes are required — additional rules are added to the rule library.

8. Dashboard Modules

8.1 Home Screen — CWBE Reservoir Module

****Purpose:****

Displays the current water reservoir fill level, estimated days until refill, a visual progress bar, and manual pump controls. This is the primary status summary tile visible immediately on the Home screen.

****Inputs:****

- `sensors.cwbeLevel` — reservoir fill level in percent (0–100)
- `sensors.cwbeStatus` — string: 'ok' | 'low' | 'critical'
- `dev.tankLiters` — current tank fill in liters
- `TANK\_MAX\_L` = 6.5 L (physical maximum)

- `TANK\_CAPACITY\_L` = 6.0 L (nominal operational capacity)

****Processing Logic:****

1. Clamp `dev.tankLiters` to range [0, 6.5].
2. Compute display level: `lvl = round(tankLiters / 6.5 × 100)`.
3. Determine CWBE status class:

cwbeStatus Display Color Class		
----- -----		
ok	Green	
low	Amber	
critical	Red	

4. Compute refill forecast (see Section 8.7 / Water Tank Module).

****Output / Displayed Information:****

Element Content	
----- -----	
Level percentage	Integer percentage, e.g., "75%"
Liters display	e.g., "4.9 L / 6.5 L"
Status badge	"Optimal" / "Low warning" / "Critical" with animated dot
Progress bar	Proportional fill, colored by status
Refill forecast	Text label: "Refill in ~N days" or "Refill today" etc.
24h trend sparkline	Historical cwbeLevel plotted as small line graph
Crop summary chip	Active crop name or "No crops configured"
Pump status chip	"Pump on" (green) or "Pump off" (muted)

****Operational States:****

State	Condition	Visual Indicator
-----	-----	-----
Normal	cwbeStatus = 'ok'	Green badge and bar
Low Warning	cwbeStatus = 'low'	Amber badge and bar
Critical	cwbeStatus = 'critical'	Red badge and bar

****User Actions:****

Touch manual adjustment buttons to update tank level. Touch pump chip to toggle pump on/off.

****Future Expansion:****

This module is labeled "CWBE Reservoir" in the UI. The CWBE calculation engine is reserved for future integration. See Section 14.

8.2 pH Module

****Purpose:****

Displays the current pH of the nutrient solution.

****Inputs:****

- `sensors.ph` — current pH value

****Processing Logic:****

Apply `getSensorInsight('ph', sensors.ph, stage)` to determine status. See Section 5.1.

****Output / Displayed Information:****

Element	Content
-----	-----
pH value	Numeric, 2 decimal places

Status label	'Stable' / 'Slightly low' / 'Critical low' / etc.	
Tile color	Green accent bar (constant; not linked to alert state in tile)	

****Operational States:****

Status Code	Display Behavior	
-----	-----	
ok	Value displayed normally	
low	Value displayed; informational	
warn	Value displayed; alert generated	

****Missing Data:****

If `lastUpdate === null`, display "-" in the value field.

****User Actions:****

None on the Home screen tile. The Agronomist panel provides detailed pH analysis and recommendations.

8.3 Water Temperature Module

****Purpose:****

Displays the current temperature of the nutrient solution.

****Inputs:****

- `sensors.waterTemp` — current temperature in °C

****Processing Logic:****

Apply `getSensorInsight('waterTemp', sensors.waterTemp, stage)`. See Section 5.2.

****Output / Displayed Information:****

Element	Content
-----	-----
Temperature value	Numeric with "°" suffix, 1 decimal
Status label	Per insight mapping

****Missing Data:****

Display "-".

8.4 Air Temperature Module

****Purpose:****

Displays the current air temperature inside the tower enclosure.

****Inputs:****

- `sensors.airTemp` — current internal air temperature in °C

****Processing Logic:****

Apply `getSensorInsight('airTemp', sensors.airTemp, stage)`. See Section 5.3.

****Output / Displayed Information:****

Element	Content
-----	-----
Temperature value	Numeric with "°" suffix, 1 decimal
Status label	Per insight mapping

8.5 Relative Humidity Module

Purpose:

Displays the current relative humidity inside the tower enclosure.

Inputs:

- `sensors.humidity` — current RH in %

Processing Logic:

Apply `getSensorInsight('humidity', sensors.humidity, stage)` where stage is from `savedIrrigation.stage`. See Section 5.4.

Output / Displayed Information:

Element	Content	
-----	-----	
RH value	Integer with "%" suffix	
Status label	Per stage-dependent insight mapping	

8.6 Water Consumption Module

Purpose:

Estimates the net daily water consumption of the growing system and projects the number of days until the reservoir requires refilling.

Inputs:

- `savedIrrigation.stage` — active growth stage
- `dev.tankLiters` — current reservoir fill in liters
- Plant count (derived from `savedPlants`)

- `GROWTH\_STAGE\_CONSUMPTION\_RANGES` — ml/day per plant per stage

****Processing Logic:****

The net daily consumption estimate is based on observed plant water uptake (ΔW) per growth stage:

Growth Stage	Consumption Range (ml/day per plant)	Center Value
Germination	1 – 3	2.0
Seedling	6 – 8	7.0
Vegetative	10 – 14	12.0
Mature	14 – 18	16.0

****Net daily consumption formula:****

```
...
netDailyL = (centerValue × plantCount) / 1000 (L/day)
...
```

If `irrigationSaved = false` or stage is undefined: `netDailyL = 0.015 L/day` (fallback default).

****Days remaining:****

```
...
daysRemaining = tankLiters / netDailyL
...
```

****Output:****

Element	Content

| Estimated daily use | mL/day (total across all plants) |
| Days remaining | Integer or decimal, used for refill forecast display |

****Operational States:****

This module does not have standalone operational states. Its output feeds the Refill Forecast display in the Water Tank Module and the Alert Engine.

8.7 Water Tank Module

****Purpose:****

Displays the refill forecast based on the water consumption estimate and the current reservoir level.

****Inputs:****

- `dev.tankLiters` — reservoir fill in liters
- `netDailyL` — from Water Consumption Module
- `CONFIG.tank.criticalPct` = 5 %
- `CONFIG.tank.soonDays` = 3 days

****Processing Logic (getRefillForecast):****

...

remaining = clamp(tankLiters, 0, 6.5)

lvl = round(remaining / 6.5 × 100)

days = remaining / netDailyL

...

Decision table:

Condition	Forecast Class	Display Text
-----	-----	-----
Irrigation not configured	pending	"No irrigation data"
netDailyL $\leq$ 0	pending	"Refill forecast unavailable"
lvl $\leq$ 5	critical	"Low reservoir"
days < 1	critical	"Refill today"
days $\leq$ 3	soon	"Refill in ~N days"
days > 3	ok	"Refill in ~N days"

****Output / Displayed Information:****

The forecast text is displayed as a label above the CWBE reservoir block on the Home screen.

****Operational States:****

Forecast Class	UI Color Indicator
-----	-----
ok	Green
soon	Amber
critical	Red
pending	Neutral/grey

8.8 Pump Module

****Purpose:****

Displays the current pump operational state and allows the operator to manually start or stop the pump.

****Inputs:****

- `sensors.pumpActive` — boolean; true when pump is running

****Processing Logic:****

The pump state is read directly from the telemetry snapshot. Manual toggle inverts the current state and sends a hardware command.

****Output / Displayed Information:****

State	Icon Color	Status Label	
-----	-----	-----	
pumpActive=true	Green	"Running · manual"	
pumpActive=false	Grey	"Idle · on schedule"	

****User Actions:****

Toggle switch on pump card. Toggle chip in CWBE hero. Both actions call `togglePump()` which inverts `sensors.pumpActive` and fires the hardware command hook.

****Operational States:****

State	Condition	
-----	-----	
Running	sensors.pumpActive = true	
Idle	sensors.pumpActive = false	

8.9 LED Module

****Purpose:****

Displays the active LED grow light schedule, allows configuration of presets and custom schedules, and displays the current operational status.

****Inputs:****

- `savedLed.on` — LED on time (HH:MM format)
- `savedLed.off` — LED off time (HH:MM format)
- `savedLed.intensity` — brightness in percent (0–100)
- `savedLed.preset` — one of: 'seedling' | 'standard' | 'bloom' | 'custom' | 'off'
- System clock (current time)

****LED Presets:****

Preset Name	ON Time	OFF Time	Intensity
-----	-----	-----	-----
Seedling	06:00	18:00	40 %
Standard	06:00	22:00	80 %
Bloom	05:00	23:00	100 %
Off	00:00	00:00	0 %
Custom	User-defined	User-defined	User-defined

****LED Status Computation (getLedStatus):****

...

if intensity === 0 OR preset === 'off' → status = 'off'

nowMins = currentHour × 60 + currentMinute

onMins = onHour × 60 + onMinute

offMins = offHour × 60 + offMinute

if offMins ≤ onMins → status = 'scheduled' (24-hour cycle or invalid)

if nowMins ≥ onMins AND nowMins < offMins → status = 'active'

else → status = 'scheduled'

...

****LED Status Display:****

Status	Badge Text	Dot Behavior	
-----	-----	-----	
active	Active	Animated pulse	
scheduled	Scheduled	Static amber	
off	Off	Static grey	

****Output / Displayed Information (summary view):****

Element	Content	
-----	-----	
Status badge	Active / Scheduled / Off	
Preset badge	Preset name (Seedling / Standard / Bloom / Custom / Off)	
ON time	HH:MM	
OFF time	HH:MM	
Daily hours	Calculated: offMins – onMins (in hours)	
Intensity	Percentage (e.g., 80%)	

****Daily hours formula:****

...

hours = max(0, (offMins – onMins) / 60)

...

****User Actions:****

"Edit LED" button opens the editor. Editor contains:

- Preset selector (Seedling / Standard / Bloom / Custom / Off)
- ON time input (HH:MM)
- OFF time input (HH:MM)
- Intensity slider (0–100%)
- Cancel and Save buttons

Selecting a preset auto-fills ON time, OFF time, and intensity. Manually adjusting any value changes preset to 'custom'. Save commits values to `savedLed`. Cancel restores to last saved values.

****Irrigation Alignment Check:****

When the LED schedule and irrigation window are both configured, the system checks alignment (see Section 8.13 and the LED alignment logic in Section 10).

8.10 Photoperiod Module

****Purpose:****

Tracks and displays the daily photoperiod duration (hours of LED-on time per day).

****Inputs:****

- `savedLed.on`, `savedLed.off`, `savedLed.intensity`, `savedLed.preset`

****Processing Logic:****

```

if intensity === 0 OR preset === 'off': photoperiodHours = 0

else: photoperiodHours = max(0, (offMins - onMins) / 60)

```

Note: Overnight schedules ($\text{offMins} \leq \text{onMins}$) are treated as "scheduled" status. The daily hours calculation in V1 does not compute wrap-around overnight hours. Overnight cycles are not typical in GL Towers operation.

****Output:****

Element	Value
-----	-----

| Daily on-hours | e.g., "16h" or "0h" |

Displayed as a statistic inside the LED summary card.

8.11 Alert Module

****Purpose:****

Displays the current active alerts. Accessible via the bell icon in the status bar. Alerts are categorized by severity and grouped by recency.

****Inputs:****

`AppState.notifications` — global alert array.

****Displayed Information:****

Each alert entry shows:

- Alert title
- Alert message
- Severity indicator (color stripe: red = critical, amber = warning, blue = info)
- Device name (if multi-device environment)
- Timestamp (relative, e.g., "2 min ago")
- Read/unread state (opacity reduced for read items)

****Alert Count Badge:****

The bell icon displays a badge with the count of unread alerts (red badge, max display typically 9+).

****User Actions:****

- Tap alert entry: marks as read
- "Clear" button: removes all alerts from the log

****Operational States:****

State	Condition
-----	-----
No alerts	Notification list is empty
Active alerts	One or more unread entries exist

8.12 Recommendation / Agronomist Module

****Purpose:****

Provides the operator with rule-based diagnostic analysis of the current system state. Displays the health status, health score, trend analysis, and actionable recommendations derived from the Agronomist Rule Engine.

****Inputs:****

- `dev.sensors` — current snapshot
- `dev.telemetryHistory.samples` — rolling sample array
- `dev.savedIrrigation.stage` — active growth stage
- `dev.savedPlants` — plant configuration
- `dev.tankLiters` — reservoir level

****Processing Logic:****

The Agronomist module calls `getAgronomistReport(dev)` which returns a complete analytical report. This function performs:

1. ****Confidence computation**** — assess reliability of the analysis based on history depth.
2. ****History context**** — compute trends for pH, waterTemp, airTemp, humidity, cwbeLevel.
3. ****Plant profile resolution**** — resolve the dominant plant's optimal parameter ranges.
4. ****Rule evaluation**** — evaluate all AGRONOMIST_RULES against current device state.

5. **Health score and status** — from fired rules (Section 6.3).
6. **Refill forecast** — from getRefillForecast() (Section 8.7).
7. **Plant scope** — multi-plant compatibility analysis.
8. **Telemetry offline detection** — if last update > 300,000 ms ago.

****Confidence Levels:****

Coverage (hours of history)	Confidence Score	Label
-----	-----	-----
< 3 samples total	0.00	None (snapshot only)
< 1 hour	0.30	Live
1 – < 6 hours	0.55	Partial history
6 – < 24 hours	0.80	Limited history
≥ 24 hours	1.00	Extended history

****Trend Computation:****

Trends are computed over the last 10 samples (trendWindow). A minimum of 3 samples is required. The last 10 samples are split in half; the mean of the second half is compared to the mean of the first half. If the difference exceeds the threshold, a trend direction is reported.

Parameter	Trend Threshold (delta to trigger)
-----	-----
pH	0.20
Water Temp	0.50 °C
Air Temp	0.50 °C
Humidity	3.00 %
CWBE Level	3.00 %

Trend values: 'rising' | 'falling' | 'stable' | 'unknown'

****Telemetry Offline Threshold:****

If ``Date.now() - sensors.lastUpdate > 300,000 ms`` (5 minutes), telemetry is considered offline. The Agronomist panel displays an offline warning and suspends crop analysis.

****Output / Displayed Information:****

Element	Content
-----	-----
Health status badge	Excellent / Good / Attention / Critical
Health score	Integer 0–100
Primary message	Summary text from highest-priority fired rule
Recommendations	Action text from highest-priority fired rule
Trend indicators	Rising / Falling / Stable per parameter
Confidence label	History coverage description
Plant profile section	Target pH, water temp, air temp, RH for dominant crop

****Telemetry Offline Display:****

When offline: "Telemetry Unavailable — Crop analysis is temporarily suspended."

****User Actions (Advanced mode):****

The Agronomist panel includes a chat interface. The operator can type questions and receive rule-based diagnostic responses. This is a text-processing module — it does not affect health evaluation or alerts.

8.13 Connectivity Module

****Purpose:****

Displays the current connection status of the tower device and signal quality.

****Inputs:****

- ``dev.metadata.connectionStatus`` — 'online' | 'warning' | 'offline'

- `dev.metadata.signalQuality` — 'excellent' | 'good' | 'fair' | 'poor'
- `dev.sensors.lastUpdate` — timestamp of last telemetry received

****Processing Logic (updateDeviceMetadataSync):****

If `metadata.overrideConnectionStatus` is set, that value is used directly.

Otherwise:

...

age = Date.now() - sensors.lastUpdate

staleThreshold = 120,000 ms (2 minutes)

if age > staleThreshold × 2 (240 s): connectionStatus = 'offline'

if age > staleThreshold (120 s): connectionStatus = 'warning'

else: connectionStatus = 'online'

...

****Output / Displayed Information:****

Status	Display
online	Live dot (green, animated)
warning	Warning indicator (amber)
offline	Offline indicator (red/grey)

The live dot is displayed in the page subtitle of the Home screen: "Sensors live · simulation mode" (or "live data" in production).

The connectivity status is also shown in:

- Device metadata panel (Profile screen → device info)

- Notification if data becomes stale (> 120 s without update)

9. Alert Engine

9.1 Architecture

The Alert Engine is separate from the UI rendering pipeline. It maintains a persistent log of state-based alerts and discrete event alerts. The UI reads from this log but does not control it.

9.2 Alert Types

****State-Based Alerts (deduplication key prevents duplicates):****

These alerts are continuously evaluated on each notification generation run. An alert persists in the log as long as its condition remains true. When the condition clears, the alert is removed from the active set.

Alert Key Pattern	Trigger Condition	Severity
reservoir-crit-{did}	Reservoir level $\leq$ 5%	Critical
refill-today-{did}	Estimated remaining autonomy < 1 day	Critical
refill-soon-{did}	Estimated remaining autonomy $\leq$ 3 days	Warning
telemetry-stale-{did}	No telemetry received for more than 120,000 ms	Critical
ph-warn-{did}	pH diagnostic status = Warning	Warning
ph-low-{did}	pH diagnostic status = Critical Low	Critical
wt-warn-{did}	Water temperature diagnostic status = Warning	Warning
at-warn-{did}	Air temperature diagnostic status = Warning	Warning
at-low-{did}	Air temperature diagnostic status = Critical Low	Critical
rh-warn-{did}	Relative Humidity diagnostic status = Warning	Warning
rh-low-{did}	Relative Humidity diagnostic status = Critical Low	Critical
irr-align-{did}	Irrigation schedule is not aligned with LED photoperiod	Warning

****Event Alerts (additive, not deduplicated by condition):****

These are one-time log entries triggered by operator actions:

Event Type	Trigger
-----	-----
refill	Manual tank level adjustment
led_saved	LED configuration saved
irr_saved	Irrigation configuration saved
device_add	New device added

9.3 Priority Rules

When multiple alerts are active simultaneously, the bell badge shows the total unread count. The alert log displays them in reverse chronological order (newest first), with new alerts inserted at the top.

The notification center groups alerts by "Today" and earlier periods.

9.4 Highest Priority Alert

For notification summary purposes, the highest-severity active alert determines the overall system status communicated to the operator. Priority order: critical > warning > info.

9.5 Throttling

Throttle Parameter	Value	Purpose
-----	-----	-----
generationThrottleMs	8,000 ms	Minimum interval between full generation runs
eventThrottleMs	10,000 ms	Suppress duplicate event alerts within window

9.6 Storage Limit

Maximum alerts stored in memory and persisted: ****50****. When the limit is reached, the oldest alerts are discarded.

9.7 User Notification Behavior

- New alert: bell badge count increments; badge visible on bell icon.
- Tapping an alert entry: marks it as read (reduced opacity in list).
- Tapping "Clear All": removes all alerts from log and resets badge.

10. Recommendation Engine

10.1 Architecture

Recommendations are generated by the Agronomist Rule Engine as part of ``getAgronomistReport()``. Each rule in ``AGRONOMIST_RULES`` carries both an explanation key (describes the condition) and a recommendation key (describes the corrective action).

10.2 Rule Library

The complete rule library with priorities, conditions, and recommendation actions:

Rule ID	Domain	Severity	Priority	Trigger Condition	Recommended Action
waterTemp.too_warm	Water Temperature	Critical	10	Water temperature $\geq 27^{\circ}\text{C}$	Move the reservoir away from heat sources. Replace approximately one-third of the reservoir water with cooler water. If the nutrient solution is old or roots show signs of stress, replace the solution completely.

waterTemp.warm	Water Temperature	Warning	11	25 °C ≤ Water temperature < 27 °C	Monitor temperature closely. If it remains above 25 °C for an extended period, replace approximately one-third of the reservoir water with cooler water.
waterTemp.slightly_warm	Water Temperature	Info	12	23 °C ≤ Water temperature < 25 °C	Continue monitoring. No immediate action required unless the temperature continues to rise over time.
ph.too_low	pH	Critical	20	pH < 5.5	Raise pH immediately using pH-Up. Recheck every 15 minutes until the pH stabilizes within the optimal range (5.8–6.5).
ph.too_high	pH	Critical	21	pH > 7.5	Lower pH immediately using pH-Down. Recheck every 15 minutes until the pH stabilizes within the optimal range (5.8–6.5).
ph.slightly_low	pH	Info	22	5.5 ≤ pH < 5.8	Monitor the pH trend. If it continues to decrease, make a small pH-Up adjustment and recheck after 30 minutes.
ph.high	pH	Warning	23	7.2 < pH ≤ 7.5	Add pH-Down in small increments. Wait approximately 30 minutes before making additional corrections.
ph.slightly_high	pH	Info	24	6.5 < pH ≤ 7.2	Continue monitoring. If the pH remains above the optimal range for an extended period, perform a small corrective adjustment.
airTemp.too_hot	Air Temperature	Critical	30	Air temperature ≥ 28 °C	Reduce ambient temperature immediately. Improve ventilation or move the tower away from heat sources.
airTemp.warm	Air Temperature	Warning	31	26 °C ≤ Air temperature < 28 °C	Improve ventilation and reduce nearby heat sources to return to the optimal operating range.
airTemp.slightly_warm	Air Temperature	Info	32	24 °C ≤ Air temperature < 26 °C	Continue monitoring. Improve air circulation if the condition persists.

airTemp.cool	Air Temperature	Warning	33	$12\text{ }^{\circ}\text{C} \leq \text{Air temperature} < 16\text{ }^{\circ}\text{C}$	Increase ambient temperature gradually to restore optimal growing conditions.
airTemp.too_cold	Air Temperature	Critical	34	Air temperature $< 12\text{ }^{\circ}\text{C}$	Increase temperature immediately by providing supplemental heating or relocating the tower to a warmer environment.
humidity.too_low	Relative Humidity	Critical	40	$\text{RH} < \text{Stage Minimum} - 10\%$	Verify reservoir level, irrigation schedule and enclosure sealing. Investigate possible air leaks.
humidity.low	Relative Humidity	Warning	41	$\text{Stage Minimum} - 10\% \leq \text{RH} < \text{Stage Minimum}$	Inspect enclosure sealing, irrigation frequency and monitor the water consumption trend.
humidity.high	Relative Humidity	Warning	42	$\text{Stage Maximum} < \text{RH} \leq 99\%$	Monitor for condensation, stagnant air and unusual odours. Ensure adequate air circulation.
humidity.too_high	Relative Humidity	Warning	43	$\text{RH} > 99\%$	Inspect for persistent condensation, stagnant air pockets and abnormal odours. Improve ventilation if required.
reservoir.critical	Reservoir	Critical	50	Refill Forecast = Critical	Refill the reservoir immediately before the next irrigation cycle.
reservoir.soon	Reservoir	Warning	51	Refill Forecast = Soon	Schedule a refill to maintain uninterrupted irrigation and system stability.

10.3 Priority Rules

Rules are sorted by ascending priority number before display. The lowest priority number fires first and is displayed as the primary observation.

10.4 Suppression Rules

- Rules with `condition()` returning false are never included in the output.

- If `dev.sensors.lastUpdate === null`, all rules return false (no telemetry → no evaluation).
- If telemetry is offline (> 300,000 ms), plant-profile and crop analysis is suspended but sensor-based rules remain active until lastUpdate = null.

10.5 Irrigation Alignment Check

The alignment check is not part of the Agronomist Rule Engine; it is an operational configuration warning:

...

getLedAlignmentStatus(savedIrrigation, savedLed):

```

if LED preset = 'off' OR intensity = 0: return 'ok'
if offMins ≤ onMins (24h LED): return 'ok'
if nightBreak = false: return 'warn'
if winStart ≥ ledOn AND winEnd ≤ ledOff: return 'ok'
else: return 'warn'
...

```

Warning text when misaligned: "Irrigation window (HH:MM–HH:MM) extends outside LED active hours (HH:MM–HH:MM)."

10.6 Substrate-Specific Cycle Warnings

These are informational warnings generated during irrigation configuration (not Agronomist rules):

Substrate	Condition	Warning Text
-----	-----	-----
Soil	cyclePause < 30 min	Soil substrates need longer dry periods. A pause of at least 30 min prevents waterlogging.
Hydroponic	cyclePause > 60 min	Hydroponic roots need frequent circulation. Pauses over 60 min may reduce oxygenation.

| Clay | cyclePause > 45 min | Clay pebbles drain fast — consider a shorter pause for more frequent cycles. |

11. Missing Data Handling

11.1 Pre-Telemetry State

When a device has been created but no telemetry snapshot has yet been received (`sensors.lastUpdate === null`), the following behavior applies:

Module	Behavior
-----	-----
Sensor tiles	Display "-" in value fields
Health Engine	Returns no status; "No Data" indicator displayed
Agronomist Engine	All rules return false; no analysis displayed
Alert Engine	State-based alerts do not fire (rules guard on lastUpdate)
Refill Forecast	Returns 'pending' if irrigation not configured
Trend Analysis	Returns 'unknown' for all parameters
Confidence Score	0.00 (none) when fewer than 3 samples available

11.2 Individual Missing Fields

If a specific sensor field is present in the snapshot but contains an invalid value (NaN, null, out-of-range), the associated module displays "-" and the corresponding rules do not fire. Other parameters with valid values continue to be evaluated normally.

12. Communication Errors

12.1 Stale Data Detection

If a telemetry snapshot has not been received within the stale threshold, the following actions occur:

Time Without Update	Action
> 120,000 ms	Connection status → 'warning'
> 240,000 ms	Connection status → 'offline'
> 120,000 ms	'telemetry-stale' alert generated (critical severity)

The displayed sensor values on the Home screen continue to show the last received values (they are not cleared). The live indicator pulse is the only real-time signal that data is current.

12.2 Telemetry Offline State for Plant Analysis

When `Date.now() - sensors.lastUpdate > 300,000 ms`:

- The Agronomist panel displays: "Telemetry Unavailable — Crop analysis is temporarily suspended."
- The last time a valid update was received is displayed as "Last update received N minutes ago."
- Sensor-based rules that already fired prior to the offline event remain in the notification log until explicitly cleared or conditions recover.

12.3 Recovery

When a new telemetry snapshot arrives after a stale or offline period:

- `lastUpdate` is refreshed.
- `connectionStatus` is recomputed to 'online'.
- All health rules are re-evaluated with the new data.
- The `'telemetry-stale'` alert is removed from the active alert set automatically on the next notification generation run.

13. Refresh Strategy

13.1 Event-Driven Updates

The dashboard does not poll the screen on a fixed timer. Screen updates occur exclusively when:

1. A telemetry snapshot is received (from the telemetry provider).
2. An operator action is performed (navigate, toggle, adjust, save, etc.).

This design minimizes unnecessary rendering cycles and is appropriate for embedded display implementation.

13.2 Telemetry Tick Intervals

Speed Setting	Tick Interval	Typical Use	
-----	-----	-----	
Slow	30,000 ms	Power-saving mode	
Normal	15,000 ms	Default operation	
Fast	5,000 ms	Active monitoring	

13.3 Notification Generation Throttle

Notifications are regenerated at most once every **8,000 ms** regardless of telemetry speed. Calling the notification generator more frequently than this threshold is a no-op.

13.4 History Sample Rate

A new history sample is recorded at most once every **10,000 ms** (regardless of telemetry speed). At the default 15,000 ms telemetry interval, one sample is recorded per tick.

13.5 Persistence

Action	Storage Write Frequency
-----	-----
Device state (tank level)	Every 4 telemetry ticks
Telemetry history	Every 4 history samples
Notifications	On every generation run
LED / irrigation configuration	On each save action

14. Reserved Future Modules

14.1 CWBE — Crop Water Balance Efficiency

****Status:** Excluded from Dashboard V1. Reserved for future integration.**

CWBE is an advanced scientific module currently under validation. It computes a water-balance efficiency metric that correlates plant water uptake with irrigation delivery to detect inefficiencies, root health indicators, and early stress signatures.

The dashboard architecture has been designed to allow future integration of CWBE without requiring structural redesign.

Reserved integration points:

Point	Location in Architecture
-----	-----
CWBE score computation	Health Engine — additional rule group, priority 60–69
CWBE display panel	Home screen CWBE hero block (currently shows reservoir)
CWBE trend analysis	Telemetry history sample (`cwbeLevel` field already recorded)
CWBE alert rules	Alert Engine — additional dedup keys

The current `cwbeLevel` field in the telemetry snapshot is a reservoir percentage placeholder that will be replaced or supplemented by the CWBE efficiency score when the module is integrated.

14.2 Correlation Engine

****Status: Reserved for future implementation.****

A cross-parameter correlation engine will identify statistical relationships between sensor readings (e.g., RH drop correlated with increased consumption). The telemetry history rolling buffer (480 samples) provides the data substrate for this analysis. No structural changes to the data model are required.

14.3 Predictive Analytics

****Status: Reserved for future implementation.****

The refill forecast (Section 8.7) is the first implementation of predictive analytics. Future extensions will include predicted pH drift, thermal envelope prediction, and growth milestone scheduling based on stage data.

14.4 AI Assistant

****Status: Reserved for future implementation.****

The Agronomist chat interface in Advanced mode currently uses rule-based pattern matching. The interface architecture is designed for replacement with an LLM-backed response engine when available. The same chat input and output display infrastructure will be reused.

14.5 Machine Learning

****Status: Reserved for future implementation.****

The telemetry history storage format, health scoring architecture, and rule priority system are designed to support future ML-based anomaly detection and scoring. All decisions remain explainable and overridable by the rule-based system in parallel.

15. Appendix

A — CONFIG Reference Table

Parameter	Value	Unit	Description
telemetry.simulationTickMs	15,000	ms	Default sensor update interval
telemetry.staleThresholdMs	120,000	ms	Threshold for stale data warning
notifications.generationThrottleMs	8,000	ms	Minimum interval between notification generation
notifications.eventThrottleMs	10,000	ms	Suppress duplicate event alerts within the configured time window
notifications.maxCount	50	count	Maximum number of stored notifications
tank.capacityL	6.0	L	Nominal operational reservoir capacity
tank.maxL	6.5	L	Maximum physical reservoir capacity
tank.pumpFlowLPerMin	0.10	L/min	Measured pump flow rate
tank.criticalPct	5	%	Reservoir level at or below this value triggers a critical alert
tank.soonDays	3	days	Remaining autonomy threshold for refill-soon notification
tank.cwbeCritPct	15	%	Reserved CWBE critical threshold (future implementation)
tank.cwbeWarnPct	30	%	Reserved CWBE warning threshold (future implementation)
sensors.ph.warnLo	5.5	pH	Lower warning threshold
sensors.ph.lo	5.8	pH	Lower optimal operating limit
sensors.ph.hi	6.5	pH	Upper optimal operating limit
sensors.ph.warnHi	7.0	pH	Upper warning threshold (display logic)
sensors.waterTemp.warnLo	16	°C	Lower warning threshold
sensors.waterTemp.lo	18	°C	Lower optimal operating limit
sensors.waterTemp.hi	22	°C	Upper optimal operating limit

sensors.waterTemp.warnHi	27	°C	Upper warning / critical threshold
sensors.airTemp.warnLo	16	°C	Lower warning threshold
sensors.airTemp.lo	18	°C	Lower optimal operating limit
sensors.airTemp.hi	26	°C	Upper optimal operating limit
sensors.airTemp.warnHi	28	°C	Upper warning threshold
agronomist.penalty.critical	40	pts	Health Score deduction for each critical rule
agronomist.penalty.warning	15	pts	Health Score deduction for each warning rule
agronomist.penalty.info	5	pts	Health Score deduction for each informational rule
agronomist.healthBand.excellent	85	pts	Minimum score for "Excellent" system status
agronomist.healthBand.good	65	pts	Minimum score for "Good" system status
agronomist.healthBand.attention	40	pts	Minimum score for "Attention" system status
agronomist.confidenceBand.live	1	hrs	Minimum telemetry coverage for "Live" confidence
agronomist.confidenceBand.partial	6	hrs	Minimum telemetry coverage for "Partial" confidence
agronomist.confidenceBand.limited	24	hrs	Minimum telemetry coverage for "Limited" confidence
agronomist.ph.optimalLo	5.8	pH	Agronomist optimal pH lower limit
agronomist.ph.optimalHi	6.5	pH	Agronomist optimal pH upper limit
agronomist.ph.toleratedHi	7.2	pH	Highest experimentally tolerated pH
agronomist.ph.attentionHi	7.5	pH	High pH attention threshold
agronomist.ph.warnLo	5.5	pH	Low pH warning threshold
agronomist.humidity.criticalLo	65	%	Critical low humidity (fallback, no growth stage)
agronomist.humidity.warnLo	80	%	Warning low humidity (fallback, no growth stage)
agronomist.humidity.warnHi	99	%	Near-saturation humidity threshold
agronomist.airTemp.criticalLo	12	°C	Critical low air temperature
agronomist.airTemp.warnLo	16	°C	Warning low air temperature
agronomist.airTemp.warnHi	26	°C	Warning high air temperature
agronomist.airTemp.criticalHi	28	°C	Critical high air temperature
agronomist.trendDelta.ph	0.20	pH	Minimum variation required to detect a pH trend
agronomist.trendDelta.waterTemp	0.50	°C	Minimum variation required to detect a water temperature trend
agronomist.trendDelta.airTemp	0.50	°C	Minimum variation required to detect an air temperature trend

agronomist.trendDelta.humidity	3.00	%	Minimum variation required to detect a humidity trend
agronomist.trendDelta.cwbeLevel	3.00	%	Reserved minimum variation for CWBE trend detection
agronomist.trendWindow	10	samples	Number of samples used for trend calculations
agronomist.minSamplesForTrend	3	samples	Minimum number of samples required before trend analysis is performed

B — Irrigation Matrix

All values: ON duration (min) / OFF duration (min). Baseline: Italy indoor, March/April, 18 °C, 0.10 L/min pump.

Growth Stage	Soil	Coco Coir	Hydroponic	Clay Pebbles	Rockwool
Germination	1 min ON / 360 min OFF	1 min ON / 300 min OFF	1 min ON / 240 min OFF	1 min ON / 210 min OFF	1 min ON / 300 min OFF
Seedling	1 min ON / 180 min OFF	1 min ON / 150 min OFF	1 min ON / 120 min OFF	1 min ON / 105 min OFF	1 min ON / 180 min OFF
Vegetative	1.5 min ON / 150 min OFF	1.5 min ON / 120 min OFF	1.5 min ON / 90 min OFF	1.5 min ON / 75 min OFF	1.5 min ON / 150 min OFF
Mature	2 min ON / 90 min OFF	2 min ON / 75 min OFF	2 min ON / 60 min OFF	2 min ON / 50 min OFF	2 min ON / 90 min OFF

C — Temperature Modifier for Irrigation Off-Cycle

Temperature Category	Air Temp Range	OFF-Cycle Multiplier
-----	-----	-----
Cool	< 18 °C	× 1.15
Normal	18 – 23 °C	× 1.00
Warm	23 – 28 °C	× 0.82
Hot	> 28 °C	× 0.67

D — Crop Demand Divisor for Irrigation Off-Cycle

Crop Category	OFF-Cycle Divisor
---------------	-------------------

--	--

Leafy greens	1.00
--------------	------

Basil	0.95
-------	------

Parsley	0.80
---------	------

Herbs	0.73
-------	------

Microgreens	0.68
-------------	------

Fruiting	1.20
----------	------

E — LED Preset Reference

Preset	ON	OFF	Intensity	Photoperiod
--------	----	-----	-----------	-------------

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Seedling	06:00	18:00	40 %	12 h
----------	-------	-------	------	------

Standard	06:00	22:00	80 %	16 h
----------	-------	-------	------	------

Bloom	05:00	23:00	100 %	18 h
-------	-------	-------	-------	------

Off	00:00	00:00	0 %	0 h
-----	-------	-------	-----	-----

F — Growth Stage Humidity Targets

Stage	RH Min	RH Max	Critical Low (tooLow = min – 10)
-------	--------	--------	----------------------------------

--	--	--	--

Germination	80 %	96 %	70 %
-------------	------	------	------

Seedling	70 %	96 %	60 %
----------	------	------	------

Vegetative	60 %	96 %	50 %
------------	------	------	------

Mature	50 %	96 %	40 %
--------	------	------	------

G — Growth Stage Net Consumption Ranges

Stage	Min (ml/day/plant)	Max (ml/day/plant)	Center (ml/day/plant)
-------	--------------------	--------------------	-----------------------

Germination	1	3	2.0
Seedling	6	8	7.0
Vegetative	10	14	12.0
Mature	14	18	16.0

H — Health Score Penalty Table

Severity	Penalty per rule	Notes	
critical	– 40 pts	Two critical rules → score ≤ 20	
warning	– 15 pts	Three warning rules → score 55	
info	– 5 pts	Ten info rules → score 50	

I — CWBE Level Thresholds (V1 — Reservoir Placeholder)

Level	Status	cwbeStatus Field
> 30 %	Normal	'ok'
15 – 30 %	Low	'low'
< 15 %	Critical	'critical'
≤ 5 % (alert)	Alert	Triggers critical notification

J — Connection Status Thresholds

Age of Last Update	connectionStatus	Behavior	
≤ 120 s	online	Live dot animated	
120 s – 240 s	warning	Warning indicator shown	
> 240 s	offline	Offline indicator shown	
> 120 s (alert trigger)	—	telemetry-stale alert generated	

| > 300 s (agronomist) | — | Plant analysis suspended |